



May 2015

Team: SC, RM, RS, LC (75%), NW (20%)

Strategy

- [LC] Research and reporting on landscape of primary mortgage market
 - Refinanceable Percentage of Mortgage Universe: Presented initial model to the Investment Committee. Continued refinements.
 - Tracking employment levels in the Mortgage Banking industry
 - Historical differences in mortgage rates between small and large lenders
- [SC] CRE Business Launch: Pitchbook, Saba Track Record

Modeling

- **Scale:** Reorganize and refactor existing CCMQuantlib fixed income analytics library (written by Bin Shao) as Scale. See the Scale presentation (distributed separately) for an overarching view of the goals of this library.
 - [JH, NW, RM] New Scale: Work on productionizing Quantlib code that replicates Bin's code. If successful, this approach would provide an excellent platform for our Scale project. Quantlib is:
 - Reasonably well-tested: it's been around since 2000
 - Well-maintained: There are at least a couple of releases every year
 - Rich feature set: Quantlib functionality includes a significant collection of utilities (Date libraries, optimization engines, splines etc.), yield curve builders, interest rate models etc.
 - Documented code
 - [RM] Isolated prepayment model from CCMQuantlib and incorporated it into the DLL project structure for Scale. The Prepayment Module is able to accept a bond name and deliver an SMM vector as output.
 - [NW] Merged recent changes in Yield Curve Builder into updated version of Scale
 - [NW] Continued to resolve debugging errors in Yield Curve Builder
- **Baton** [RM]: Agency prepayment model
 - Completed work on curtailment module. Presented to Investment Committee.



Analytics

- **Ensemble** [RM, JH]: Middle-tier – allows for decoupling Scale from front-end applications
 - Expose functions in Scale for front-end users to test/use the Scale library



June 2015 Pipeline

Team: SC, RM, LC (75%), NW (50%)

Strategy

- [LC] Research and reporting on landscape of primary mortgage market
 - Wrap-up calculation of refinancable universe
- [LC, RM] Develop a proprietary Mortgage Credit Availability (MCA) measure by running the most recent originations through Accordion each month and tracking several measures of delinquency and default.
- [SC] Refinement of IOS Basis Model
- [SC] Research on primary mortgage market
- [SC] CRE Strategy: Document Investment Process and approach to Risk Management

Modeling

- **Scale** [NW]: Complete fixing bugs in Yield Curve Builder. Hook up Yield Curve Builder with Ensemble. Improve logging/error-handling for Yield Curve Builder, refactor code to improve readability and structure.
- **Accordion** [RM, LC]: Roll-rate model for GSE loan-level data
 - Productionize running CRT/STACR deals with eMBS CRT data on a monthly basis
 - Generate a daily forecast of HPA/mortgage rates for the base case.
 - Generate daily delinquency/default vectors for CRT deals using stress scenarios from Moody's/Economy.com and a base case.
 - Generate daily yield numbers for various CRT deals and tranches using these delinquency/default vectors.
 - Create a monthly surveillance report based on the CRT monthly files
- **Baton** [RM]: Agency prepayment-model
 - Default and Turnover modules